

8. The Duffing Oscillator

The Duffing oscillator is one of the prototype systems of nonlinear dynamics. It first became popular for studying anharmonic oscillations and, later, chaotic nonlinear dynamics in the wake of early studies by the engineer Georg Duffing [8.1]. The system has been successfully used to model a variety of physical processes such as stiffening springs, beam buckling, nonlinear electronic circuits, superconducting Josephson parametric amplifiers, and ionization waves in plasmas. Despite the simplicity of the Duffing oscillator, the dynamical behavior is extremely rich and research is still going on today.

8.1 The Duffing Equation

The Duffing oscillator is described by the differential equation

$$\ddot{x} + r\dot{x} + \omega_0^2 x + \beta x^3 = f \cos \omega t, \quad (8.1)$$

which differs from the elementary textbook example of a forced and damped harmonic oscillator only by the nonlinear term βx^3 , which changes the dynamics of the system drastically. The *harmonic oscillator*

$$\ddot{x} + r\dot{x} + \omega_0^2 x = f \cos \omega t \quad (8.2)$$

offers, however, a convenient starting point for a subsequent discussion of the nonlinear case. For $\beta = 0$, a solution is given by

$$x(t) = A \cos(\omega t - \phi), \quad (8.3)$$

which appears to be an ellipse

$$\left(\frac{x(t)}{A}\right)^2 + \left(\frac{v(t)}{\omega A}\right)^2 = 1 \quad (8.4)$$

in phase space $(x, v) = (x, \dot{x})$. The amplitude A and phase shift ϕ depend on the driving frequency ω , which shows typical resonance behavior when ω is varied:

$$A(\omega) = \frac{f}{\sqrt{(\omega_0^2 - \omega^2)^2 + (r\omega)^2}} \quad (8.5)$$

$$\tan \phi(\omega) = \frac{r\omega}{\omega_0^2 - \omega^2}. \quad (8.6)$$

The resonance frequency is

$$\omega_{\text{res}} = \sqrt{\omega_0^2 - (r/2)^2}. \quad (8.7)$$

The general solution of the equation of motion is a superposition of the solution (8.3) and the general solution of the homogeneous equation, i.e. it can be written as

$$x(t) = A \sin(\omega t + \phi) + c_+ e^{\lambda_+ t} + c_- e^{\lambda_- t}, \quad (8.8)$$

where the constants $c_{\pm}$ are determined by the initial conditions $x(0)$, $\dot{x}(0)$, and

$$\lambda_{\pm} = -r/2 \pm \sqrt{(r/2)^2 - \omega_0^2}. \quad (8.9)$$

The solution of the inhomogeneous differential equation (8.3) is the unique limit cycle. For $r > 0$, the contribution of the homogeneous equation vanishes in the long time limit and each solution $x(t)$ approaches (8.3). It should be recalled that the two regions can be distinguished: in the underdamped case $r < 2\omega_0$, the approach to the limit cycle is slow and oscillatory, while in the overdamped case, $r > 2\omega_0$, it is slow and exponential. Both regions are separated by the aperiodic limiting case $r = 2\omega_0$, where the limit cycle is reached in the shortest time. For decreasing values of the force amplitude f , the amplitude A goes to zero and the limit cycle — the ellipse (8.4) — shrinks to a limit point. In a stroboscopic picture (i.e. using a flashlight at times $0, T, 2T, 3T, \dots$, where $T = 2\pi/\omega$ is the period of the excitation frequency) the limit cycle appears as a single point in phase space, which appears at

$$(x, v) = (A \cos \phi, -A\omega \sin \phi). \quad (8.10)$$

It should be noted that such a stroboscopic view can be considered as a Poincaré section. This discussion exhausts the possible dynamical behavior of the pure harmonic case.

The appearance of the nonlinear term βx^3 in the Duffing equation (8.1) changes the situation completely:

- An analytic solution is no longer available.
- The superposition principle is no longer valid. A linear combination of solutions does not solve the equation of motion.
- The qualitative behavior is infinitely richer:
 There appear coexistent limit cycles of different periods and the long time limit of the motion is sensitively dependent on the initial conditions.
 One observes an entirely different type of long time behavior, where the trajectory approaches a strange attractor.
 Different strange attractors and limit cycles can coexist. Their basins of